Module: AI Analyst and assistant

English version of ../../modules/ai-analyst.md.

1. Purpose and boundaries

Answers questions about the data through tools rather than direct database access: backend/app/ai/tools/{analytics,customers,scenarios,actions}.py. The model provider is replaceable (ai/providers/base.py, implementations ollama.py, mock.py). It does not execute arbitrary SQL, does not reach data bypassing the tools, and does not work when AEZA_AI_ENABLED=0.

2. Use scenarios

- The question "which services are at risk" → a tool call → an answer with numbers and sources.
- An answer without a model (mock provider) in tests and with AI disabled.
- Failure: the provider is unavailable → a clear error, regular calculations keep working.

3. Contract

/api/v1/ai/*, /api/v1/model/* in backend/openapi.json. The internal contract is ai/service.py and the schemas in ai/schemas.py; tools declare their input and output explicitly.

4. Data

No tables of its own: reading goes through tools that call the analytics domain. Answers are not stored as a source of truth — numbers always come from the domain.

5. Access rules

Authentication is required; the tools are available only within the analyst role. Input and output safety checks live in ai/safety.py (escaping, context limits).

6. Dependencies

Outgoing: the analytics domain, the LLM provider over HTTP, the RAG index (ai/rag/index.py). Incoming: backend/app/api/routes_ai.py.

7. Configuration

AEZA_AI_ENABLED, AEZA_AI_PROVIDER, AEZA_AI_MODEL, AEZA_AI_BASE_URL, AEZA_AI_MAX_TOKENS, AEZA_AI_TIMEOUT_SECONDS, AEZA_LLM_*, AEZA_OLLAMA_*. Disabled by default: AI is an option, not a condition of the product working.

8. Build and debug

```
python -m pytest backend/tests/test_ai.py -q
python -m backend.app.ai.rag --docs ./docs --out backend/data/rag_index.json
```

9. Tests

`backend/tests/test_ai.py`: tool calls, the disabled mode, the mock provider, protection against unsafe output. The "AI is not a source of truth" check — numbers in the answers match the domain.

10. Observability

The provider state is visible through `/api/v1/model/*`; provider errors are logged without prompt contents. Latency metrics — `/metrics`; the default timeout of 420 s shows up for the operator as long requests.

11. Limitations and readiness

- Answer quality is not guaranteed; it is compensated by taking the numbers from the domain.
- The RAG index is built into the image at build time — after changing `docs/` a rebuild is required.
- Readiness: with AI disabled the product is fully functional; the tools cannot expose data the user is not allowed to see.